Diverse Cohort outcomes explainer

reag: Modified NIA Reagan, per thresholds set by Rush:

High likelihood that dementia is due to Alzheimer's disease lesions:
CERAD: Definite/frequent/C3 and Braak: Stage ≥ V

Intermediate likelihood
CERAD: Definite/frequent/C3 and Braak: Stage ≥ I
OR:
CERAD: Probable/moderate/C2 and Braak: Stage ≥ III

Low likelihood:
CERAD: Probable/moderate/C2 and Braak: ≤ II
OR:
CERAD: Possible/sparse or infrequent/C1 and Braak: any
OR:
CERAD: No AD and Braak: Stage ≥ I

No AD:
CERAD: No AD/None/C0 and Braak: Stage 0

MayoDx:
Mayo criteria (NINCDS-ADRD criteria as determined by a neurologist)**:**
AD: Braak ≥ 4, plaque observations quantified (in absence of non-AD tauopathy, Thal ≥ 2)
Control: Braak ≤ 3, no/minimal plaque (Thal < 2)
Other: all other

Mayo Dx, subset of Mayo patients from UFL cohort: these diagnoses follow NIA guidelines for neuropathological diagnosis of AD (Hyman et al., PMID: 22265587 and Montine et al., PMID: 22101365). AD case assigned to patients with any degree of AD neuropathological diagnosis.

ADoutcome (Derived outcome):

AD, Control, Other categories: derived harmonized outcomes based on CERAD neuritic plaque score and Braak stage, or NINCDS-ADRD criteria for Mayo. ***The following criteria seek to harmonize outcomes between cohorts as closely as possible, with Mayo's neuropath-based diagnoses (as determined by Mayo specialists) set as the standard to match***. These thresholds will also exclude individuals with age-related tauopathies from being classified as AD cases or controls.

Rush, MSSM, Columbia derived outcome criteria:
AD: Braak ≥ 4, CERAD = Frequent or Moderate
Control: Braak ≤ 3, CERAD = Sparse or None
Other: all other

UFL (from Mayo): some patients from UFL were not consistent with other MayoDx criteria (i.e. 8 individuals had Braak < 4 and a mayoDx==AD). These individuals were assigned 'Other' criteria in the ADoutcome variable to harmonize more closely with MayoDx criteria.